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The Price of De-correlation: Quantifying the Agreement–Cost Trade-off in Heterogeneous Large Language Model Ensembles

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ABSTRACT Large language model (LLM) ensembles are increasingly used as decision-making committees under the assumption that aggregating multiple agents improves reliability. However, ensembles composed of repeated instances of the same model may produce correlated judgments and create an appearance of consensus without genuine independence. This study quantifies the relationship between cross-provider model heterogeneity, ensemble agreement, and inference cost. In a 54,000-call primary experiment, we evaluate three five-agent ensemble configurations with increasing provider diversity on a directional financial-analysis task covering 100 equities across 12 dates. Within the evaluated configurations, greater cross-provider heterogeneity is associated with significantly lower within-ensemble agreement, with a primary reduction of $\Delta\kappa = 0.336$ (95% CI: 0.304–0.369), but increases inference cost by approximately $4.4\times$. Because agreement does not by itself measure dependence between errors, we additionally estimate the correlation between agents' binary error indicators. Error correlation decreases by $\Delta\phi = 0.454$ (95% CI: 0.376 to 0.534), while mean per-agent accuracy remains essentially unchanged. This indicates that the observed change reflects lower error dependence rather than differences in average correctness. The contrast remains substantial under approximate capability matching, although that control does not isolate provider from other model-specific characteristics. We also introduce an independent-draw measurement protocol designed to prevent shared sampling conditions from inflating estimated agreement. A pre-registered replication on a substantially different task involving 537 expert clinical-decision items reproduces the directional pattern. Agreement decreases by $\Delta\kappa = 0.072$ (95% CI: 0.040 to 0.104), while error correlation decreases by $\Delta\phi = 0.393$ (95% CI: 0.325 to 0.459). Although heterogeneity does not produce a statistically reliable improvement in average directional accuracy, exploratory analysis shows that homogeneous ensembles generate unanimous incorrect decisions at 1.84 times the rate of fully heterogeneous ensembles (95% CI: 1.42 to 2.52). Taken together, the results show that heterogeneous model composition can reduce error dependence without guaranteeing greater predictive accuracy, while imposing a measurable inference cost premium.

INDEX TERMS Ensemble learning, error correlation, large language models, model heterogeneity, multi-agent systems.

I. INTRODUCTION

MULTI-AGENT large language model systems increasingly aggregate multiple agents through voting, debate, or other consensus mechanisms [1], [2]. These approaches are commonly motivated by the wisdom-of-crowds principle, under which combining multiple judgments can reduce individual errors. Their effectiveness, however, depends on the extent to which the participating agents produce independent errors. An ensemble comprising repeated instances of the same model may exhibit substantial output correlation, thereby providing less effective diversity than its number of agents suggests. Cross-provider model heterogeneity offers a potential remedy, but it also introduces additional inference cost, and its ability to reduce correlated behavior has not

been established quantitatively. Recent evidence that LLMs may exhibit correlated errors even across providers and architectures further raises concerns about algorithmic monoculture [3].

Existing research supports two competing expectations regarding cross-provider heterogeneity. Under the diversity hypothesis, models developed by different providers, using distinct training procedures, datasets, and architectures, should generate less-correlated outputs and therefore contribute more diverse judgments. In contrast, the algorithmic-monoculture hypothesis [3] suggests that common training data, objectives, and evaluation practices may cause substantial behavioral similarity even across model families. These perspectives imply materially different agreement–cost relationships. If provider identity contributes little additional diversity, increasing heterogeneity will raise inference cost without meaningfully reducing agreement. Conversely, a substantial reduction in agreement would indicate that cross-provider diversity is measurable and can be evaluated against its associated cost. This study tests these alternatives empirically.

Despite the growing use of heterogeneous LLM ensembles, three related issues remain insufficiently examined. First, few studies directly quantify the extent to which cross-provider heterogeneity is associated with lower within-ensemble agreement. Second, the corresponding inference-cost premium is rarely incorporated into evaluations of ensemble design. Third, agreement estimates may be distorted when supposedly independent agents share random seeds, sampling trajectories, cached outputs, or other elements of the generation context. Consequently, existing evaluations may conflate genuine model similarity with artifacts of the measurement procedure. We address these limitations by jointly estimating the reduction in ensemble agreement, the associated inference cost, and the effect of independent-draw measurement controls.

This distinction is practically important because LLM ensembles are increasingly considered for financial analysis and other consequential decision-support applications in which multiple agents are expected to provide independent second opinions. When agents exhibit highly correlated behavior, repeated calls may increase computational expenditure while producing an overstated impression of evidential consensus. Practitioners therefore require evidence on both the degree of independence obtained through model heterogeneity and the cost required to obtain it. By estimating these quantities jointly, this study provides an empirical basis for selecting ensemble configurations according to the reliability and cost requirements of a given application.

a: Contributions

This paper makes the following contributions.

1) *Pre-registered empirical measurement.* We conduct a pre-registered empirical study comprising 54,000 model calls to quantify the association between cross-provider heterogeneity and within-ensemble agreement. Agreement decreases substantially and monotonically as provider diversity increases, with a primary contrast of $\Delta\kappa = 0.336$ and a 95% confidence interval of $[0.304, 0.369]$.
2) *An empirical agreement–cost frontier.* We quantify the inference-cost premium associated with cross-provider diversity. Moving from the homogeneous to the fully heterogeneous configuration reduces agreement at an approximately $4.4\times$ increase in inference cost.
3) *A reproducible independent-draw measurement protocol.* We separate genuine model agreement from agreement induced by shared sampling conditions. In a controlled pilot, shared per-run seeding inflated the estimated agreement contrast by approximately fourfold, whereas unique per-agent seeds removed the artifact. To support replication, we provide a frozen, hash-verified dataset, fixed experimental configurations, and a deterministic seeded analysis pipeline that regenerates the reported figures and tables.
4) *An exploratory analysis of consensus structure.* Although heterogeneity does not produce a statistically reliable improvement in average directional accuracy, homogeneous ensembles generate unanimous incorrect directional decisions at $1.84\times$ the rate of fully heterogeneous ensembles, motivating confirmatory evaluation of collective-error risk.

We distinguish reduction in ensemble agreement from improvement in predictive accuracy. Cross-provider heterogeneity substantially increases observed behavioral diversity, but it does not reliably improve average correctness in the evaluated directional-finance task. Thus, model diversity should be evaluated according to whether the additional diversity improves the downstream objective rather than being treated as beneficial by default.

II. RELATED WORK

A. LLM ENSEMBLES AND MIXTURE-OF-AGENTS

Aggregating multiple LLM outputs through voting, debate, or layered mixture-of-agents architectures has been investigated as a means of improving response quality and reliability [1], [2], [4]. However, prior studies indicate that naive aggregation may provide only limited gains over the strongest individual model and may amplify systematic errors when correlated weaker models outvote a stronger model [3]. Related scaling studies also report diminishing or negative returns from repeatedly adding homogeneous agents, whereas more diverse ensembles can provide more sustained improvements [5].

These findings have motivated diversity-aware methods such as the focal-diversity pruning strategy used in LLM-TOPLA [6], the fingerprint-based ensemble selection of DFPE [7], and heterogeneous consensus frameworks such as Council Mode [8]. Such approaches generally evaluate diversity through downstream performance. In contrast, the present study treats within-ensemble agreement as the primary quantity of interest and directly measures how agree-

ment, statistical uncertainty, and inference cost vary across configurations with differing cross-provider heterogeneity.

B. CORRELATED ERRORS AND ALGORITHMIC MONOCULTURE

Kim *et al.* [3] document substantial error correlation across a large set of LLMs, including models developed by different providers, and relate this dependence to the systemic risk of algorithmic monoculture. Their analysis concerns error correlation—that is, the tendency of models to fail on the same instances—whereas the present study examines output agreement across all decisions, regardless of whether those decisions are correct.

These quantities are related but not equivalent. Lower output agreement may indicate greater behavioral diversity, but it does not necessarily imply less-correlated errors or improved predictive performance. In the vision-language setting, Bugaud [9] similarly reports that models from the same architectural family exhibit correlated errors that reduce the effective number of independent voters and shows that family-aware aggregation can improve ensemble performance.

The present work extends this line of inquiry to text-based LLM committees by measuring provider-level agreement, comparing same-provider and cross-provider agent pairs, and incorporating the inference-cost trade-off associated with model heterogeneity.

C. INTER-RATER AGREEMENT AND DIVERSITY METRICS

Fleiss' κ [10] is a chance-corrected agreement statistic designed for categorical ratings produced by multiple raters. We apply it by treating the five LLM agents as interchangeable raters, each $(\text{ticker}, \text{date})$ pair as a rated subject, and BUY, HOLD, and SELL as nominal categories.

Unlike diversity measures used primarily to select, weight, or prune ensemble members [6], our objective is to estimate agreement itself and determine how it varies with provider heterogeneity and inference cost. Because Fleiss' κ can be sensitive to category prevalence, we additionally assess the robustness of the observed configuration ordering using Krippendorff's α [11] and Gwet's AC1 [12]. Statistical uncertainty is estimated through cluster-bootstrap resampling at the equity level [13].

D. REPRODUCIBILITY AND PRE-REGISTRATION

Pre-registration and frozen, hash-verified datasets remain uncommon in empirical LLM research. We adopt a confirmatory design—specifying the analysis plan in advance, collecting data once, and re-analyzing deterministically—to limit researcher degrees of freedom, and we release the resulting artifacts. The freezing, seeding, and reproducibility procedures are described in detail in Section IV-D.

III. PROBLEM FORMULATION AND METRICS

TABLE 1. Model registry (verified via a live probe). Prices are USD per 10^6 tokens at time of experiment.

Provider (role)	Model string	In	Out
OpenAI (primary)	gpt-5.4-nano	0.20	1.25
Google (het.)	gemini-3.5-flash	1.50	9.00
xAI (het.)	grok-4.3	1.25	2.50

A. ENSEMBLE CONFIGURATIONS

Each ensemble contains five agents that independently respond to the same prompt through separate application programming interface (API) calls. We designate OpenAI as the primary provider and define the heterogeneity level of configuration c as the number of agents drawn from a non-primary provider:

$$h_c = \sum_{i=1}^{5} \mathbb{I}\left(p_i \neq p_{\text{primary}}\right), \tag{1}$$

where p_i denotes the provider of agent i. We evaluate three configurations:

- **HOM** (h=0): five agents using the primary-provider model;
- **HET-LITE** (h=1): four primary-provider agents and one Google agent;
- **HET** (h=3): two primary-provider agents, two Google agents, and one xAI agent.

The exact model identifiers, provider roles, and token prices used during data collection are reported in Table 1.

B. AGREEMENT STATISTIC

For each configuration c, experimental run r, and evaluation cell $s = (\text{ticker}, \text{date})$, the five agents return one categorical direction from $\mathcal{Y} = \{\text{BUY}, \text{HOLD}, \text{SELL}\}$. We treat agents as raters and evaluation cells as rated subjects. Fleiss' κ is computed separately within each run, producing the run-level agreement statistic $\kappa_{c,r}$. Configuration-level agreement is then defined as the mean across the $R = 3$ runs:

$$\kappa_c = \frac{1}{R} \sum_{r=1}^{R} \kappa_{c,r}. \tag{2}$$

The pre-registered primary endpoint is the reduction in agreement between the homogeneous and fully heterogeneous configurations:

$$\Delta\kappa_{\text{HOM}-\text{HET}} = \kappa_{\text{HOM}} - \kappa_{\text{HET}}. \tag{3}$$

The pre-registered secondary contrasts are

$$\Delta\kappa_{\text{HOM}-\text{HET-LITE}} = \kappa_{\text{HOM}} - \kappa_{\text{HET-LITE}},$$
$$\Delta\kappa_{\text{HET-LITE}-\text{HET}} = \kappa_{\text{HET-LITE}} - \kappa_{\text{HET}}.$$

Positive values indicate that configurations with greater cross-provider heterogeneity show lower within-ensemble agreement.

C. INDEPENDENT-DRAW ESTIMAND

The estimand of interest is agreement among independently sampled agent responses. Accordingly, every agent response is generated through a separate, stateless API request, without shared conversation state, response caching, or multi-sample batching, so that no two agents share a sampling trajectory through the request structure itself. Each request is additionally assigned a unique, reproducible seed:

$$\text{seed}_{r,c,s,i} = \text{SHA-256}(\text{master}, r, c, \text{ticker}, \text{date}, i) \bmod 2^{31}, \tag{4}$$

where i identifies the agent position within the ensemble. Including the agent identifier gives repeated instances of the same model distinct seeds—and therefore distinct sampling trajectories on endpoints that honor the seed parameter—while keeping those calls reproducible. All three provider endpoints used in this study accept a seed parameter, and the unique per-agent seed is transmitted on every call, masked to a positive 32-bit integer as some endpoints require; the model registry and supplementary manifest record the seed sent with each call. Because provider seed support is best-effort rather than a determinism guarantee, reproducibility ultimately rests on the archived raw responses (Section IV-D) in addition to re-executable seeding.

As a diagnostic for residual within-run dependence, we compare agreement among the five draws generated within a run with agreement estimated from independently generated draws across runs. For the homogeneous configuration, these quantities should be approximately equal when all five agents are sampled independently. A materially positive within-run minus cross-run gap would indicate that shared seeds or another common sampling condition is artificially increasing measured agreement. For heterogeneous configurations, the cross-run calculation also introduces additional same-model replicate pairs and is therefore interpreted as a secondary consistency check rather than a direct independence test.

D. UNCERTAINTY QUANTIFICATION

Uncertainty in the agreement contrasts is estimated using a cluster bootstrap with equities as the resampling unit [13]. Clustering at the equity level preserves the dependence among repeated analysis dates for the same security while allowing variation across securities to determine the confidence interval. For each of 2,000 bootstrap replications, equities are sampled with replacement, all associated dates and agent responses are retained, and the configuration-level agreement statistics and contrasts are recomputed. We use the percentile method with a fixed random-number-generator seed of 42 to construct 95% confidence intervals (CIs). Under the pre-registered decision rule, the primary hypothesis is considered supported when $\Delta\kappa_{\text{HOM}-\text{HET}} > 0$ and its 95% confidence interval excludes zero.

Equity is the pre-registered resampling unit because repeated observations are nested within securities. Each equity is evaluated on 12 analysis dates, and observations for the same security share company-specific context. Analysis dates constitute a second crossed source of dependence because all securities evaluated on a given date share common market conditions.

Because equity and analysis date are crossed rather than nested, we additionally assess uncertainty under a two-way resampling procedure. This follows the crossed-resampling principle of the pigeonhole bootstrap [14], in which the two margins are resampled separately, and is motivated by the broader multiway-inference literature for non-nested clustering dimensions [15]; it is not an implementation of a multiway variance estimator. We therefore assess the primary contrast under three resampling schemes: equity clustering over 100 equities, date clustering over 12 dates, and a two-way procedure in which equities and dates are resampled independently, with multiplicity preserved on both margins, and their sampled intersections retained. Results appear in Table 5. The point estimate is unchanged across schemes. The two-way interval is the widest, as expected when both dependence structures are admitted, and it continues to exclude zero.

Because the date dimension contains only 12 clusters, inference based on that margin may be imprecise. The pre-registered equity-clustered interval therefore remains the primary analysis, while the date-clustered and two-way results are reported as sensitivity analyses.

IV. EXPERIMENTAL DESIGN

A. TASK AND GRID

For each evaluation cell $s = (j, t)$, where j denotes an equity and t denotes an analysis date, every agent receives the same dated information set and independently returns a response conforming to the following JavaScript Object Notation (JSON) schema:

```
{
  "direction":  "BUY | HOLD | SELL",
  "conviction": <integer 1-5>,
  "rationale":  "<free-text explanation>"
}
```

The `direction` field constitutes the categorical output used in the primary agreement analysis. The `conviction` field is an integer from 1 to 5, and the `conviction` and `rationale` fields are retained for secondary and exploratory analyses. The complete prompt template and response schema are frozen and included in the reproducibility package.

Each response is parsed as strict JSON against this schema. A response that fails to parse, omits a required field, or returns a direction outside $\{\text{BUY}, \text{HOLD}, \text{SELL}\}$ is treated as invalid and reissued with the same per-agent seed, up to a pre-registered maximum of five retries, with every attempt logged. Calls interrupted by provider daily-quota limits were deferred and re-issued after the quota reset under the same seeds. A cell that remained invalid after the retry budget would have been handled under the pre-registered exclusion rule; in the confirmatory dataset, all 54,000 grid cells ultimately returned schema-valid responses, so no cells were

excluded. The complete attempt history, including failed and deferred calls, is preserved in the released log.

The experiment covers 100 large-cap equities distributed across all 11 Global Industry Classification Standard sectors and 12 analysis dates in 2026. For each equity-date pair, responses are generated under three ensemble configurations, three independent runs, and five agents per ensemble. The complete design therefore contains $100 \times 12 \times 3 \times 3 \times 5 = 54{,}000$ individual model calls.

The dates are selected to reduce the possibility that model responses rely on memorized post-event information and to ensure that the 20-trading-day forward-return window used in the secondary accuracy analysis has fully elapsed. They are also chosen to span distinct 2026 market conditions. The equity universe, sector allocation, inclusion criteria, and complete ticker list are reported in the supplementary materials.

B. NO-LOOKAHEAD CONTEXT

Each agent receives a price-derived information set constructed solely from observations available through the trading day immediately preceding the corresponding analysis date. For every equity-date cell, the context contains the trailing one-week return, the trailing one-month return, the trailing three-month return, the 21-trading-day realized volatility, and the security's position within its trailing 63-trading-day price range. No observations from the analysis date or any subsequent date are included in the agent prompt. The same information set is supplied to all agents evaluating a given cell, ensuring that differences in responses arise from model behavior rather than differences in the available inputs.

For exact replication, the return horizons are defined in trading days. For example, the range-position feature is computed as

$$\mathrm{RangePos}_{j,t} = \frac{P_{j,t-1} - \min_{\tau \in W_{63}} P_{j,\tau}}{\max_{\tau \in W_{63}} P_{j,\tau} - \min_{\tau \in W_{63}} P_{j,\tau}}, \qquad (5)$$

where W_{63} denotes the preceding 63-trading-day window. End-of-day prices are obtained from a single market-data vendor (Financial Modeling Prep), fetched once, and frozen in the released cache; realized volatility is computed over the trailing 21-trading-day window, and the treatment of missing observations is specified in the supplementary materials. Data provenance and the exact information supplied to each model call are recorded in the experimental logs.

C. MODELS

All model calls are executed using a sampling temperature of 0.7. Exact provider and model identifiers are fixed for the duration of data collection and reported in Table 1, together with their roles in each ensemble configuration and the input- and output-token prices applicable during the experiment. Each agent response is generated through a separate API request under the independent-sampling protocol described in Section III-C. Apart from the model and provider assignments defined by the ensemble configuration, agents evaluating the same cell receive identical prompts and inference settings.

TABLE 2. Primary and secondary ′κ (cluster bootstrap over equities, 2,000 draws, seed 42).

Contrast	$\Delta\kappa$	95% CI
HOM − HET (primary)	0.336	[0.304, 0.369]
HOM − HET-LITE	0.249	[0.218, 0.280]
HET-LITE − HET	0.087	[0.067, 0.105]

Because proprietary model endpoints may be modified or replaced by their providers, the archived call-level dataset constitutes the empirical record of the experiment. The conclusions therefore apply to the responses generated by the specified endpoints during the reported collection period rather than to future versions carrying similar product names. To support replication and auditability, the experimental log records the model identifier, provider, timestamp, seed, prompt hash, token usage, cost, and raw response for every call.

D. PRE-REGISTRATION, FREEZING, AND REPRODUCIBILITY

Empirical evaluations of LLM systems involve several design choices, including model selection, prompt construction, sampling parameters, evaluation endpoints, exclusion rules, and statistical procedures. Revising these choices after observing the results can increase researcher degrees of freedom and make replication more difficult. We therefore adopt a confirmatory workflow that separates pre-specified analyses from subsequent exploratory investigations.

Before confirmatory data collection, we froze the hypotheses, ensemble configurations, prompt template, experimental grid, random-seed construction, primary and secondary endpoints, uncertainty-estimation procedure, and decision rule. The resulting pre-registration was assigned a SHA-256 hash to establish that the design preceded the collection of the confirmatory dataset. The released package contains timestamped cryptographic freeze receipts establishing that the pre-registered design preceded data collection. Each receipt records the UTC freeze time, the commit that carried the frozen artifacts, and the SHA-256 of the configuration, agent and analysis code, prompt template, pre-registration, grid, and model manifest; the cross-domain replication has its own receipt, recorded before any API call for that arm. Raw outputs were collected once and stored in an immutable, read-only, hash-verified dataset. The analysis pipeline reads only the frozen dataset and experimental configuration, and all stochastic procedures, including bootstrap resampling, use fixed seeds. The pipeline regenerates the reported tables and figures deterministically.

Analyses not included in the original pre-registration are identified explicitly as exploratory or as documented amendments. This separation prevents post hoc results from being presented as confirmatory evidence and allows the primary findings to be evaluated against the design specified before data collection. The frozen dataset, pre-registration materials,

orchestration configuration, and analysis code are included in the accompanying reproducibility package.

E. COLLECTION PROVENANCE AND RETRY ACCOUNTING

The confirmatory grid contains 54,000 planned responses, all of which ultimately returned valid outputs. Completing the grid required 94,566 API requests. Of the 40,566 unsuccessful requests, 40,544 were provider rate-limit or quota responses and 22 were transport or schema failures. Because rate-limit and quota responses returned no model completion, retrying them did not select observations based on generated content.

Every request attempt is retained in the released log. Request settings were fixed across configurations, including temperature 0.7, per-agent seed construction, a 256-token completion limit, and strict JSON output. The released reproducibility materials contain the prompts, prompt hashes, model identifiers, prices, experimental configuration, and preregistration records.

V. RESULTS

The analysis includes the complete planned set of 54,000 model calls across 100 equities, 12 analysis dates, three ensemble configurations, three independent runs, and five agents per ensemble. Unless otherwise stated, agreement statistics are computed separately within each run and then averaged across runs.

A. HETEROGENEITY AND ENSEMBLE AGREEMENT (PRIMARY)

Within-ensemble agreement decreases monotonically as cross-provider heterogeneity increases. Fleiss' κ is 0.552 for HOM, 0.302 for HET-LITE, and 0.215 for HET, as reported in Table 4 and illustrated in Fig. 1. Using the unrounded configuration-level estimates, the pre-registered primary contrast is

$$\Delta\kappa_{\mathrm{HOM-HET}} = \kappa_{\mathrm{HOM}} - \kappa_{\mathrm{HET}} = 0.336,$$

with a 95% cluster-bootstrap confidence interval of $[0.304, 0.369]$. Because the confidence interval excludes zero, the pre-registered primary hypothesis that greater cross-provider heterogeneity is associated with lower within-ensemble agreement is supported. In Landis–Koch [16] terms, homogeneous ensembles reach "moderate–substantial" internal agreement while fully heterogeneous ones fall to "fair." Both pre-registered secondary contrasts are also positive, with confidence intervals excluding zero (Table 2); under the two-way scheme of Section III-D they remain so, at $[0.208, 0.290]$ and $[0.045, 0.123]$ respectively. Agreement therefore decreases at each observed increase in heterogeneity, from HOM to HET-LITE and from HET-LITE to HET. Across the evaluated configurations the ordering is monotonic: agreement is lower at each higher level of cross-provider heterogeneity in this task.

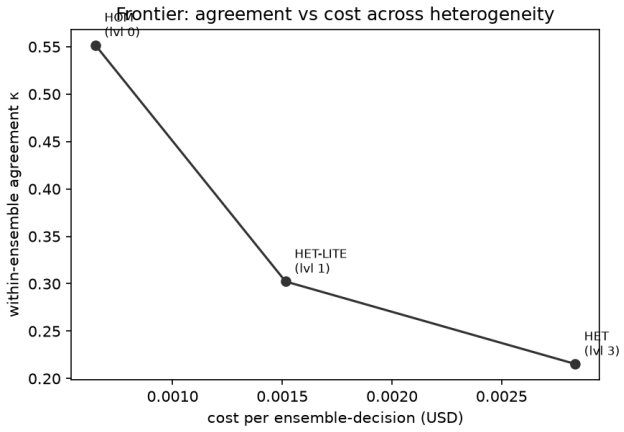


FIGURE 1. Agreement–cost frontier. Within-ensemble Fleiss' κ (y) versus cost per ensemble decision (x) at heterogeneity levels $\{0, 1, 3\}$. Lower κ indicates lower observed within-ensemble agreement; the horizontal spread is the cost premium.

TABLE 3. Agreement under three statistics (per-run averaged), with the per-config directional marginals. All three statistics agree on the ordering despite the prevalence shift in HOLD.

	agreement			marginals		
Config	κ	α	AC1	BUY	HOLD	SELL
HOM	0.552	0.552	0.743	0.212	0.709	0.079
HET-LITE	0.302	0.303	0.548	0.282	0.653	0.064
HET	0.215	0.216	0.398	0.358	0.556	0.085

TABLE 4. Frontier: within-ensemble agreement and cost by heterogeneity level.

Config	h	κ	\$ / ensemble-decision
HOM	0	0.552	6.48×10^{-4}
HET-LITE	1	0.302	1.52×10^{-3}
HET	3	0.215	2.84×10^{-3}

B. ROBUSTNESS OF THE AGREEMENT STATISTIC

The marginal distribution of BUY, HOLD, and SELL responses changes across ensemble configurations. In particular, the proportion of HOLD responses decreases from 0.71 in HOM to 0.56 in HET. Because Fleiss' κ can be sensitive to category prevalence (the "κ paradox"), this shift could affect its absolute value independently of changes in the underlying level of agreement. We therefore recompute agreement using Krippendorff's α [11] for nominal outcomes and Gwet's AC1 [12]. All three statistics produce the same monotonic configuration ordering: HOM > HET-LITE > HET (Table 3). Krippendorff's α closely tracks Fleiss' κ, while Gwet's AC1 produces higher absolute values but preserves both the ordering and the substantial HOM-to-HET reduction. The primary conclusion is therefore robust to alternative treatments of chance agreement and is not attributable solely to changes in the category marginals.

TABLE 5. Primary contrast $\Delta\kappa_{\mathrm{HOM-HET}}$ under alternative resampling units. The point estimate does not depend on the resampling scheme.

Resampling unit	Resampled units	$\Delta\kappa$	95% CI
Equity (pre-reg.)	100 equities	0.3363	[0.3035, 0.3689]
Analysis date	12 dates	0.3363	[0.2989, 0.3656]
Two-way	100×12	0.3363	[0.2786, 0.3877]

C. AGREEMENT, ERROR CORRELATION, AND DECORRELATION

Agreement and error dependence capture different properties of an ensemble. We therefore distinguish five concepts used throughout the analysis:

1) *Output agreement*: the extent to which agents return the same categorical label. This is measured by Fleiss' κ and constitutes the primary agreement endpoint.
2) *Error correlation*: dependence between agents' mistakes. This is measured using ϕ computed from binary error indicators.
3) *Sampling independence*: whether repeated responses share elements of the sampling process, such as a random seed or decoding trajectory.
4) *Model diversity*: differences in model weights, architecture, training data, or related model characteristics.
5) *Provider diversity*: whether agents are supplied by different organizations. The ensemble configurations vary provider composition, although provider remains bundled with other model-specific characteristics.

In this study, decorrelation denotes a reduction in error correlation, measured by ϕ. The agreement measure κ captures a related but distinct change in categorical consensus. Accordingly, the ''price of de-correlation'' in the title refers to the additional inference cost associated with ensemble configurations that exhibit lower error dependence.

Pairwise dependence between binary error indicators is a standard characterization of classifier-ensemble diversity, and that literature also shows that diversity and ensemble accuracy are related but not interchangeable [17]. We therefore complement the categorical agreement statistic κ with a ϕ correlation computed from paired error indicators. Within each evaluation cell, that is each (configuration, equity, date, run) combination, we form for every unordered pair of agent positions the pair of binary error indicators defined against the realized forward-return sign. These paired indicators are pooled over all cells and all agent-position pairs within a configuration, and ϕ is the Pearson correlation of that pooled set, so one ϕ is reported per configuration rather than one per agent pair. Error correlation decreases from $\phi = 0.984$ for HOM to 0.742 for HET-LITE and 0.530 for HET, giving $\Delta\phi_{\mathrm{HOM-HET}} = 0.454$ (95% CI $[0.376, 0.534]$). Under the two-way scheme that also admits date dependence, the interval widens to $[0.328, 0.586]$ and still excludes zero.

Because treating HOLD as an abstention changes the scoreable subset, we repeat the analysis with HOLD counted as an error. The corresponding values are 0.586, 0.359, and 0.277, with $\Delta\phi_{\mathrm{HOM-HET}} = 0.309$ (95% CI $[0.262, 0.354]$; two-way $[0.227, 0.387]$). The ordering is preserved under both treatments of HOLD, and under both resampling schemes.

Mean per-agent error rates are similar across configurations, indicating that the decline in ϕ reflects lower error dependence rather than differences in average correctness. The reduction is partial rather than complete, as heterogeneous agents continue to make correlated errors.

D. THE AGREEMENT–COST FRONTIER

The reduction in agreement is accompanied by higher inference cost. Cost per ensemble decision is calculated from metered prompt and completion tokens. For each call, provider-reported token counts are priced using the per-million token rates reported in Table 1. Costs are summed across the five agents within a cell and then averaged across cells within each configuration.

Prices are fixed at those applicable during data collection, so differences across configurations reflect token usage and model composition rather than later price changes. No discount, batching, or caching adjustment is applied. Failed requests that return no billable completion are excluded. The observed cost per five-agent ensemble decision increases from approximately $\$6.5 \times 10^{-4}$ for HOM to $\$1.5 \times 10^{-3}$ for HET-LITE and $\$2.8 \times 10^{-3}$ for HET. Moving from the homogeneous to the fully heterogeneous configuration therefore increases inference cost by approximately $4.4\times$ (Table 4). Fig. 1 presents within-ensemble agreement against inference cost for the three configurations. The resulting frontier quantifies the additional expenditure associated with lower agreement and allows ensemble configurations to be compared jointly on behavioral diversity and cost. The frontier does not, by itself, establish that the additional disagreement improves downstream predictive performance.

E. SHARED-SEED DEPENDENCE AND MEASUREMENT VALIDATION

a: Exactly what was shared

In the initial pilot implementation, the random seed was derived at the ensemble-run level rather than at the individual-agent level. Consequently, all five same-model agents evaluating a given cell received the same prompt and seed. Under this configuration, the calls produced near-verbatim responses, indicating that they did not function as independent draws for the purpose of measuring homogeneous-model agreement. The confirmatory implementation instead assigns every agent a unique, reproducible seed determined by its run, configuration, equity, date, and agent position (Fig. 2). The prompt, model, temperature, and experimental grid remain unchanged. This modification preserves reproducibility while allowing repeated instances of the same model to follow distinct sampling trajectories.

```
# Before (artifact): one seed shared by
# all agents in a run
seed = f(master, run)
for agent in 1..5:
    call(model, prompt, seed)  # identical draw

# After (fix): distinct per-agent seed
seed = SHA256(master, run, config,
              ticker, date, agent) & 0x7FFFFFFF
call(model, prompt, seed)  # distinct seed
```

FIGURE 2. **The one-line fix. The pilot derived the seed per run, so all five same-model agents shared a sampling path. The study assigns a distinct reproducible seed to each agent, removing the shared-seed artifact observed in the pilot. As Section III-C notes, this does not establish statistical independence of the responses.**

b: Ruling out adjacent explanations

The pilot duplication was not attributable to request batching, because every agent response was generated through a separate API request. Prompt caching was also disabled during collection. Within the controlled pilot setup, the disappearance of near-duplicate behavior after introducing per-agent seeds is therefore consistent with shared-seed dependence as the source of the inflated agreement.

c: Diagnostic verification

We evaluate residual within-run dependence by comparing within-run and cross-run agreement (Table 6). After adopting unique per-agent seeds, the HOM within-minus-cross difference is approximately 0.001, indicating that agreement among the five agents within a run is nearly identical to agreement obtained from separately generated draws across runs. In the controlled pilot, the estimated primary contrast was 0.485 under shared per-run seeding and 0.113 under independent per-agent seeding; shared seeding therefore overstated the estimated agreement contrast by approximately fourfold (Fig. 3). The pilot grid comprised 8 equities $\times$ 2 analysis dates $\times$ 2 configurations (HOM and HET) $\times$ 3 runs $\times$ 5 agents, i.e. 480 calls per seeding arm—far smaller than the confirmatory grid—so the pilot contrasts are point estimates whose role is diagnostic, demonstrating the artifact and its removal, rather than confirmatory. This factor is specific to the pilot configuration and should not be interpreted as a universal inflation constant for all LLM ensemble studies. The heterogeneous configurations produce within-minus-cross differences of -0.048 and -0.074. These differences arise because cross-run pooling introduces additional repeated draws from the same underlying models, increasing pooled agreement relative to the single-draw within-run calculation. The primary endpoint uses the within-run estimate and is therefore not increased by this pooling effect.

d: Sensitivity: agreement by pair type

We additionally separate pairwise agreement according to whether two agents use the same model. Same-model agreement is 0.800 in HOM, 0.794 in HET-LITE, and 0.862 in HET, while different-model agreement is 0.455 in HET-LITE and 0.490 in HET.

The reduction in ensemble agreement is therefore associated with the introduction of different-model pairs rather than a systematic decline in agreement among repeated instances of the same model. This analysis is a sensitivity check on the frozen dataset and does not independently verify provider seed behavior.

TABLE 6. **Non-determinism check: within-run vs. cross-run κ (must be $\approx$ 0 for the primary/homogeneous configuration).**

Config	within	cross	within$-$cross
HOM	0.552	0.551	$+0.001$
HET-LITE	0.302	0.350	-0.048
HET	0.215	0.289	-0.074

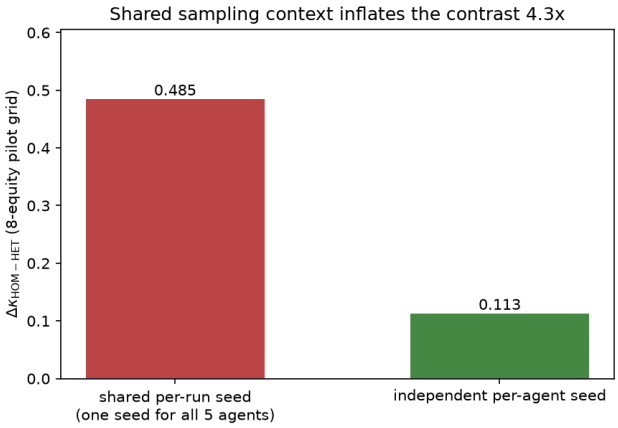


FIGURE 3. **Effect of seeding protocol on the pilot HOM$-$HET agreement contrast. The contrast is 0.485 under a shared seed and 0.113 under distinct per-agent seeds. Both estimates use the same pilot grid and model set.**

e: What unique seeds do and do not establish

Assigning a distinct seed to each agent removes the specific shared-seed artifact observed in the pilot. It does not establish statistical independence of the resulting responses.

Seed support in commercial inference systems is best-effort rather than a guarantee of deterministic execution. Distinct per-agent seeds are therefore treated as a control against one demonstrated dependence mechanism, not as proof of independence.

F. PROVIDER-PAIR AGREEMENT

The pairwise directional-agreement matrix in Fig. 4 provides additional evidence on where the reduction in ensemble agreement comes from. Same-provider agreement is 0.92 for Google agent pairs and 0.80 for OpenAI agent pairs, whereas cross-provider agreement ranges from 0.43 to 0.63. Agents supplied by the same provider therefore agree more frequently than agents supplied by different providers under identical prompt and information conditions.

This comparison should be interpreted as a configuration-specific observational finding rather than as an isolated

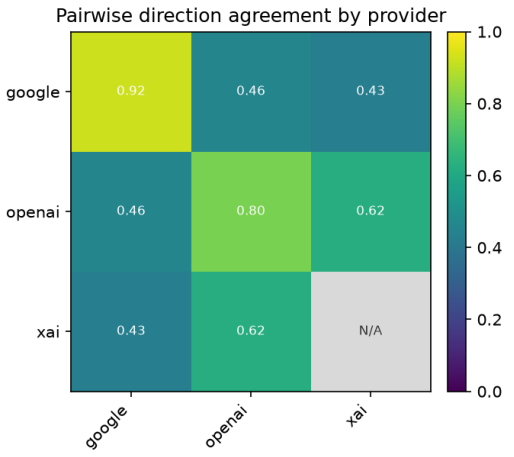


FIGURE 4. Pairwise directional-agreement matrix by provider. Diagonal (within-provider) agreement greatly exceeds off-diagonal (cross-provider), which is where the reduction in ensemble agreement originates. The xAI–xAI diagonal is undefined (marked N/A, not 0): each HET cell contains a single xAI agent, so no within-xAI agent pair exists.

provider effect. In the present design, provider is not experimentally separable from model identity, parameter count, training corpus, architecture, or alignment procedure because each provider contributes one model. The observed contrast therefore reflects substitution among the specific model and provider combinations evaluated here. The term "provider" should be understood as shorthand for that bundle of characteristics rather than as an independently identified causal factor.

Section V-J provides an additional capability-matched comparison on a second domain, in which one same-provider pair and one cross-provider pair have nearly identical member accuracy. The provider-associated ordering persists under that comparison, but provider remains inseparable from other model-specific characteristics. A design that varied model identity within provider and provider within capability tier would be required to separate these effects further.

G. SECONDARY DIRECTIONAL-ACCURACY ANALYSIS

As a pre-declared secondary analysis, we compare each ensemble's majority-vote direction with the sign of the realized 20-trading-day forward return. HOLD decisions are treated as abstentions and excluded from the corresponding hit-rate calculation. The estimated directional hit rates are 0.438 for HOM, 0.463 for HET-LITE, and 0.488 for HET (Table 7). The associated Wilson confidence intervals [18] are wide and overlapping. Moreover, the experiment was powered for the agreement endpoint rather than for differences in directional accuracy. The estimates therefore do not provide sufficient evidence that cross-provider heterogeneity improves average directional accuracy. The numerical upward trend is reported descriptively and should not be interpreted as a demonstrated performance improvement.

Hit rates at or modestly below 0.5 are the expected baseline for this task rather than a defect of the design: 20-trading-day directional forecasting from public price history alone is approximately unpredictable under near-efficient prices, and the abstention rule further shifts the effective base rate by scoring only the subset of cells on which an ensemble commits to a direction. Crucially, the agreement estimand is well-defined independent of task difficulty: Fleiss' κ and the $\Delta\kappa$ contrasts are functionals of the joint distribution of agent outputs, computed without reference to outcome labels, so the agreement results measure how similarly the agents behave regardless of whether the task admits exploitable predictive signal. A low-signal task is, if anything, a conservative setting for the primary question, because agents cannot inflate measured agreement merely by all being correct.

TABLE 7. Accuracy proxy (SECONDARY; underpowered by design). Hit-rate of majority-vote direction vs. 20-day forward-return sign; HOLD abstains.

Config	hit-rate	n	95% CI (Wilson)
HOM	0.438	315	$[0.384,\ 0.493]$
HET-LITE	0.463	339	$[0.411,\ 0.516]$
HET	0.488	525	$[0.445,\ 0.530]$

H. EXPLORATORY ANALYSIS OF CONSENSUS ERRORS

Average directional accuracy may conceal differences in how errors are distributed across committee members. We therefore conduct a post hoc exploratory analysis of consensus structure using the same frozen dataset. This analysis was not included in the pre-registered hypotheses, and its results are treated as hypothesis-generating rather than confirmatory. We examine three quantities: the proportion of decisions for which all five agents select the same label; the proportion that are unanimous, directional, and incorrect—the "manufactured-consensus" failure that monoculture warnings describe [3]; and, conditional on an incorrect majority decision, the proportion of agents endorsing that incorrect majority (Table 8).

The unanimity rate decreases from 0.591 for HOM to 0.315 for HET-LITE and 0.267 for HET. Homogeneous ensembles therefore produce identical five-agent decisions more than twice as frequently as fully heterogeneous ensembles. The proportion of decisions that are unanimous, directional, and incorrect is 0.058 for HOM, 0.030 for HET-LITE, and 0.032 for HET. The homogeneous ensemble therefore produces this form of collective error at approximately $1.84\times$ the rate of the fully heterogeneous ensemble. The equity-clustered bootstrap gives a 95% confidence interval of $[1.42,\ 2.52]$ for this ratio, which excludes unity. Because this statement is inferential rather than descriptive, we also admit date dependence under the two-way scheme used for the primary endpoint: the interval widens to $[1.13,\ 3.06]$ and still excludes unity. HET-LITE exhibits a similar reduction in unanimous directional errors at an approximately $2.3\times$ cost premium, compared with $4.4\times$ for HET.

These results are exploratory. Neither the configuration comparison nor the reported statistic was pre-registered, mul-

TABLE 8. Consensus structure by heterogeneity (exploratory; 3,600 scoreable five-agent decisions per config). "Unan. wrong" = all five agents agree on one directional call that is wrong; "pile-on" = *P*(agent endorses majority | majority wrong).

Config	unanimity	unan. wrong	pile-on
HOM	0.591	0.058	0.824
HET-LITE	0.315	0.030	0.761
HET	0.267	0.032	0.636

tiple related comparisons were available within this analysis, and no multiplicity adjustment is applied. The corresponding intervals should therefore not be interpreted as confirmatory evidence. We do not present HET-LITE as a recommended ensemble design. Its apparent cost efficiency is instead treated as a hypothesis for prospective evaluation in a confirmatory study.

Conditional on an incorrect majority vote, the proportion of agents endorsing the incorrect majority decreases from 0.824 for HOM to 0.761 for HET-LITE and 0.636 for HET. A paired equity-clustered bootstrap over the same 100 equities, using 2,000 draws and the resampling scheme of the primary endpoint, gives a HOM−HET difference of 0.189 with a 95% confidence interval of $[0.160,\ 0.216]$. When dependence across analysis dates is also included using the same two-way resampling procedure applied to the primary endpoint, the interval widens to $[0.140,\ 0.232]$ and remains above zero. Thus, within the evaluated data, incorrect majority decisions in heterogeneous ensembles are accompanied by more within-committee dissent than in the homogeneous configuration. This result remains exploratory and does not by itself establish a reduction in downstream decision risk.

Self-reported conviction does not exhibit a reliable positive relationship with per-agent directional accuracy in any configuration (Fig. 5). Higher-conviction responses do not consistently achieve higher hit rates, and no conviction category produces a reliably above-chance result. Consequently, the exploratory benefit of heterogeneity appears in the structure of collective agreement and dissent rather than in the calibration of individual agents' reported conviction. These findings motivate a separate, appropriately powered evaluation of calibration and collective-error risk.

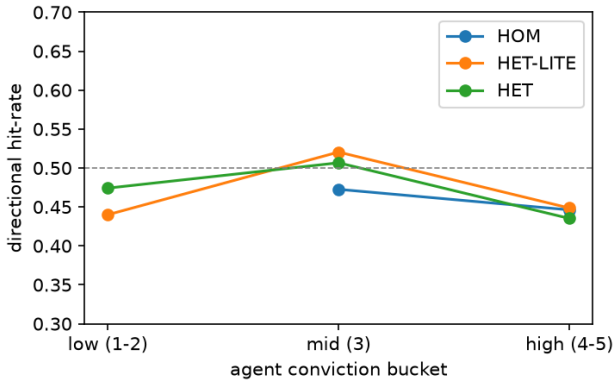


FIGURE 5. Self-reported conviction does not predict directional accuracy. Per-agent directional hit-rate by self-reported conviction bucket; no configuration rises reliably above chance (0.5, dashed), consistent with the secondary directional-accuracy analysis.

I. SELECTIVE PREDICTION

Reduced agreement is only useful if it buys something downstream, so we ask whether the configurations differ in their ability to rank their own errors. Following the selective-classification framework of El-Yaniv and Wiener [19], we evaluate the risk–coverage trade-off obtained by ranking ensemble decisions according to mean agent conviction. A decision is the majority vote of the five agents in an (equity, date) cell on the financial grid. Decisions whose majority is HOLD are excluded, as are cells whose forward-return sign is undefined or zero, which leaves the directional decisions that can be scored; the denominator for coverage is that retained set, not the full grid. Each decision is scored as erroneous when its direction disagrees with the realized forward-return sign, and is assigned the mean conviction its five agents reported. Decisions are ranked by that mean conviction in descending order, coverage is swept from the single most confident decision to the full set, and selective risk at a coverage level is the error rate among the decisions retained at that level. AURC is the trapezoidal area under the resulting risk–coverage curve, so lower is better. The HOM−HET contrast is formed within each bootstrap draw rather than between arms, since both configurations are measured on the same equities, and its interval is the equity-clustered bootstrap used for the primary endpoint, with the two-way scheme reported alongside it.

We evaluate whether reduced agreement improves selective prediction by ranking ensemble decisions according to mean agent conviction. AURC is 0.553 for HOM, 0.568 for HET-LITE, and 0.554 for HET. The HOM−HET difference is -0.001 (95% CI $[-0.059,\ 0.060]$). The corresponding two-way interval is $[-0.113,\ 0.115]$. Both intervals contain zero.

Restricting evaluation to the most confident half of decisions does not materially reduce selective risk in any configuration. These results therefore do not establish an improvement in calibration or abstention quality. The agreement and cost frontier of Section V-D describes the cost associated with greater behavioral diversity, not a demonstrated improvement in decision quality.

J. CROSS-DOMAIN REPLICATION: CLINICAL KNOWLEDGE

To test whether the observed ordering is specific to the financial task, we repeated the protocol using 537 multiple-choice items from the MMLU professional-medicine and clinical-knowledge subsets [20]. Each item presents a medical or clinical question, often as a short patient scenario describing symptoms, medical history, examination findings, or test results, and asks the model to identify the most appropriate diagnosis, treatment, next clinical step, or factual medical answer. Four candidate answers are provided and

one is designated correct by the benchmark. For example, an item may describe a patient with a particular set of symptoms and laboratory findings and ask which diagnosis or management option is most appropriate. The four response options correspond to alternative diagnoses, treatments, or clinical actions. Each of the five agents independently selects one of the four choices. Agreement is therefore measured across the agents' selected answers, while error correlation is measured by whether agents are incorrect on the same items. Choice order is shuffled deterministically to remove positional imbalance in the published answer key. The five-agent structure, three configurations, seeding procedure, temperature, number of runs, and estimators remain unchanged. The replication design, predicted direction, and decision rule were registered before collection. Confidence intervals for the clinical replication are obtained by resampling MMLU items with replacement, since the equity and analysis-date clustering structure of the financial experiment does not apply to this task.

Agreement decreases from $\kappa = 0.881$ for HOM to 0.809 for HET, giving $\Delta\kappa_{\mathrm{HOM-HET}} = 0.072$ (95% CI $[0.040, 0.104]$), compared with 0.336 in the financial experiment. Error correlation decreases by $\Delta\phi_{\mathrm{HOM-HET}} = 0.393$ (95% CI $[0.325, 0.459]$), compared with 0.454 in finance. The agreement contrast therefore changes substantially more across the two tasks than the error correlation contrast.

The pooled clinical comparison still combines provider and capability differences. However, one same-provider pair and one cross-provider pair have nearly identical member accuracy, 0.951 and 0.955 respectively. Their error correlations are 0.954 and 0.607, a difference of 0.347 (95% CI $[0.207, 0.507]$). The provider-associated ordering therefore persists after approximate capability matching, although provider remains confounded with other model-specific characteristics.

The cross-domain replication varies domain and response structure while intentionally retaining the five-agent ensemble, prompt structure, aggregation rule, and evaluated model mixtures. It therefore tests whether the observed ordering is confined to the financial task, rather than establishing invariance across ensemble-design choices. The replication supports the ordering across two substantially different domains; it does not establish that heterogeneity improves predictive accuracy, and the agreement contrast is substantially smaller in the clinical arm, so the magnitude is task-dependent.

K. ROBUSTNESS: CAPABILITY-MATCHED CONTROL

The primary ensemble configurations vary both provider identity and model capability tier. To reduce this confounding, we conduct a pre-registered capability-matched control using three OpenAI efficient-tier agents and two Google efficient-tier agents (Het-SameTier). An equivalent efficient-tier xAI model was not available, so this control varies provider across two rather than three model families. Across the same 100-equity and 12-date grid, the homogeneous baseline produces $\kappa_{\mathrm{HOM}} = 0.552$, whereas the capability-matched heterogeneous configuration produces $\kappa_{\mathrm{HET\text{-}SameTier}} = 0.236$. The resulting contrast is

$$\Delta\kappa_{\mathrm{HOM-SameTier}} = 0.315, \quad 95\%\ \mathrm{CI}\ [0.288,\ 0.341],$$

with a two-way interval of $[0.270,\ 0.361]$; both exclude zero. This capability-matched estimate is approximately 94% of the primary HOM-to-HET contrast. The result indicates that the observed reduction in agreement is largely preserved when model tier is held approximately constant. The contrast therefore remains substantial under approximate capability matching, although that control does not isolate provider from other model-specific characteristics, which are not independently identified by this design. The capability-matched control therefore reduces, but does not eliminate, the capability-tier confound, because matching is approximate and covers only two providers. The close reproduction of κ_{HOM} across independently collected datasets provides an additional consistency check for the homogeneous baseline.

L. ROBUSTNESS: TEMPERATURE SENSITIVITY

Agreement can vary with sampling temperature, raising the possibility that the primary result is specific to the main operating point of $T = 0.7$. We therefore evaluated a separate stratified subgrid comprising eight equities, two analysis dates, three ensemble configurations, and three independent runs at $T \in \{0.0, 0.7, 1.0\}$. These observations were stored separately from the frozen confirmatory dataset and analyzed using the same seeded estimator. The ordering $\kappa_{\mathrm{HOM}} > \kappa_{\mathrm{HET\text{-}LITE}} > \kappa_{\mathrm{HET}}$ is preserved at every tested temperature. The corresponding HOM-to-HET contrasts are 0.299, 0.509, and 0.385 at $T = 0.0$, 0.7, and 1.0, respectively, with 95% confidence intervals excluding zero in each case (Table 9). Absolute agreement levels, however, vary non-monotonically with temperature; for example, κ_{HOM} ranges from approximately 0.33 to 0.67. This variation reflects changes in both within-cell concordance and the marginal distribution of BUY, HOLD, and SELL responses. At $T = 0$, the homogeneous configuration produces a more nearly uniform label distribution, whereas at $T = 0.7$ and $T = 1.0$, responses are more concentrated in HOLD. The results therefore indicate that the relative reduction in agreement associated with provider heterogeneity is robust within the tested subgrid, while absolute agreement remains temperature-dependent. We note that the subgrid's contrast at the main operating point (0.509 at $T = 0.7$) differs from the full-grid estimate (0.336) at the same temperature; this discrepancy is attributable to the sampling variance of the small 8×2 subgrid, whose wide confidence interval ($[0.21,\ 0.78]$) comfortably contains the full-grid estimate. The subgrid is accordingly informative about the sign and ordering of the effect, not its magnitude. Its intervals are clustered on the eight equities only: the subgrid spans two analysis dates, which is too few clusters to support date-clustered or two-way inference, so this analysis contributes no independent multiway evidence and the intervals below should be read as indicative rather than as a test. Because this analysis uses a substantially smaller grid than the primary

TABLE 9. Temperature-sensitivity robustness (8 × 2 subgrid, 3 runs per cell). The ordering ($\kappa_{\text{HOM}} > \kappa_{\text{HET-LITE}} > \kappa_{\text{HET}}$) and the primary contrast $\kappa_{\text{HOM}-\text{HET}}$ (positive, CI excludes 0) hold at every temperature; absolute κ levels vary non-monotonically, so it is the contrast, not the absolute agreement, that is temperature-stable.

T	κ_{HOM}	$\kappa_{\text{HET-LITE}}$	κ_{HET}	$\Delta\kappa$ [95% CI]
0.0	0.33	0.17	0.03	0.30 [0.16, 0.44]
0.7	0.67	0.38	0.16	0.51 [0.21, 0.78]
1.0	0.53	0.27	0.15	0.38 [0.19, 0.54]

experiment, it is interpreted as supporting evidence rather than an independent replication. The temperature analysis was conducted under a documented protocol amendment and does not alter the frozen confirmatory dataset.

M. RELATION TO EXISTING ENSEMBLE METHODS

This research paper does not propose a new ensemble algorithm, so a direct performance benchmark against existing ensemble methods is not the primary comparison. Methods such as mixture-of-agents, multi-agent debate, and diversity-based ensemble selection are generally evaluated using downstream accuracy or win rate [2]. The present study instead measures how ensemble composition changes output agreement, error dependence, and inference cost under fixed task conditions.

The distinction is one of estimand rather than of quality. Mixture-of-agents, debate, pruning, and ensemble-selection studies ask whether a particular aggregation or selection method improves downstream performance, and a leaderboard comparison is the appropriate test of that question. This study holds the decision task, prompt, information set, and aggregation rule fixed and asks how ensemble *composition* relates to output agreement, error dependence, and inference cost. A leaderboard-style accuracy comparison would therefore answer a different question than the one posed here.

The appropriate comparison with prior work is methodological, along four axes: how diversity or agreement is measured, whether error dependence is separated from categorical agreement, whether sampling artifacts are controlled, and whether uncertainty and inference cost are quantified. The classifier-ensemble literature already establishes that diversity and ensemble accuracy are related but not interchangeable [17], which is why reduced agreement is not treated here as evidence of improved accuracy. The contribution is therefore a measurement framework that separates output agreement from error correlation, controls the shared-seed artifact observed in repeated model sampling, quantifies uncertainty under crossed dependence, and incorporates inference cost. These measurements complement performance-based evaluations rather than constitute a competing aggregation method.

VI. DISCUSSION

The results provide a quantitative characterization of the relationship between provider heterogeneity, ensemble agreement, and inference cost. Within-ensemble agreement decreases from $\kappa = 0.552$ for the homogeneous configuration to $\kappa = 0.215$ for the fully heterogeneous configuration, while inference cost increases by approximately $4.4\times$. The provider-pair analysis further shows that agents from the same provider agree more frequently than agents from different providers under identical prompt and information conditions. Together with the capability-matched analysis, these findings show that the provider-associated ordering persists across the evaluated comparisons, although provider remains confounded with other model-specific characteristics.

Reduced agreement is a necessary but not sufficient condition for improved ensemble performance. Greater behavioral diversity is useful only when the additional judgments provide relevant information or enable more reliable downstream decisions. In the present study, configurations with greater cross-provider heterogeneity show substantially lower within-ensemble agreement but no statistically reliable improvement in average directional accuracy. This result is consistent with the algorithmic-monoculture literature [3] in showing that greater behavioral diversity does not automatically translate into greater correctness. The exploratory consensus analysis nevertheless indicates that heterogeneity changes the structure of collective errors. Homogeneous ensembles produce unanimous incorrect directional decisions at approximately $1.84\times$ the rate of fully heterogeneous ensembles and exhibit greater concentration of agents around an incorrect majority decision. These findings suggest that disagreement may provide value through reduced error concentration and increased visibility of dissent, even when mean accuracy does not improve. However, this interpretation remains exploratory and requires confirmatory evaluation using an appropriately powered downstream endpoint. The same holds for selective prediction. AURC is 0.553 for HOM, 0.568 for HET-LITE, and 0.554 for HET, and the HOM$-$HET difference of -0.001 (95% CI $[-0.059, 0.060]$) contains zero. Reduced agreement did not improve error ranking, and HET-LITE, the highest AURC of the three, gains no selective-prediction advantage.

The agreement–cost frontier provides a practical framework for ensemble selection. Practitioners should not assume that the additional cost of provider heterogeneity is justified solely because it reduces agreement. Instead, they should evaluate whether the resulting diversity improves a task-relevant objective, such as calibration, robustness, abstention quality, or the frequency of concentrated collective errors. In the exploratory analysis, HET-LITE achieves a reduction in unanimous directional errors comparable to HET at an approximately $2.3\times$ cost premium, compared with $4.4\times$ for HET. Because this pattern was identified post hoc, it should be interpreted as hypothesis-generating rather than as deployment guidance. It motivates prospective evaluation of HET-LITE in a confirmatory design specifically powered to assess the tradeoff between collective error structure and inference cost.

The shared-seed pilot also has a methodological implication for empirical studies of LLM ensembles. When repeated

model instances receive the same seed or otherwise share elements of the sampling trajectory, their outputs may not represent independent draws, causing within-ensemble agreement and configuration contrasts to be materially distorted. Agreement studies should therefore document the sampling context, assign separate reproducible seeds to individual agents when supported, and include a diagnostic for residual within-run dependence.

VII. THREATS TO VALIDITY

Heterogeneity vs. capability confound. The primary ensemble configurations differ in both provider composition and model capability tier. HOM contains five instances of an efficient-tier OpenAI model, whereas the heterogeneous configurations introduce models from other providers, including higher-tier models. Consequently, part of the observed reduction in agreement could reflect capability differences rather than provider identity alone. Two analyses reduce, but do not eliminate, this concern. First, the provider-pair matrix (Fig. 4) compares agents evaluating the same equity-date cell under identical prompt and information conditions and shows substantially higher agreement for same-provider pairs (0.80–0.92) than for cross-provider pairs (0.43–0.63). Second, the capability-matched control (Section V-K) holds model tier approximately constant across OpenAI and Google and produces a contrast of $\Delta\kappa = 0.315$, approximately 94% of the primary HOM-to-HET effect. The reduction in agreement therefore remains substantial after approximate tier matching. However, the control covers only two providers because an equivalent efficient-tier xAI model was unavailable, and differences in training data, architecture, alignment procedures, and effective capability cannot be completely separated from provider identity. No model-level random-effects analysis and no fully tier-matched three-provider design were performed. The capability-matched control therefore narrows this confound rather than resolving it, and the provider-associated ordering should be read with that limitation in view.

Scope and generalizability. The primary experiment evaluates a directional financial-analysis task with a three-class BUY, HOLD, and SELL output space, while Section V-J provides a pre-registered replication on 537 expert clinical-decision items with a four-option response structure. The ordering reproduces across these two substantially different tasks, indicating that the result is not confined to the original financial setting. Its magnitude, however, differs materially across domains, with the agreement contrast in the clinical replication approximately one fifth of that observed in finance.

Several design dimensions remain fixed across the study, including prompt format, five-agent ensemble size, majority-vote aggregation, and the particular model and provider mixtures evaluated. Other task classes, model combinations, ensemble sizes, prompting strategies, or aggregation rules may produce different levels of agreement, error dependence, and cost. The present results therefore support replication of the observed ordering across the two tested domains without implying that its magnitude is invariant across settings.

Model-version pinning. Proprietary model endpoints may be updated by their providers without preserving identical behavior under the same product name. We mitigate this limitation by recording the exact model identifiers, timestamps, seeds, prompt hashes, and raw responses associated with every call and by treating the frozen, hash-verified dataset as the empirical object of record. The findings therefore apply to the archived outputs generated during the reported collection period and may not reproduce exactly with later endpoint versions.

Single-run collection. Although the design contains three independent runs per configuration, the confirmatory responses were collected within a single experimental period. Provider-side updates, transient service conditions, or undocumented inference changes occurring outside that period are therefore not averaged across multiple collection windows. Repeating the experiment at later dates would be necessary to evaluate temporal stability.

Accuracy underpowered. The experiment was designed and powered for the agreement endpoint rather than for differences in directional accuracy. The accuracy analysis uses a descriptive proxy based on the sign of the 20-trading-day forward return, treats HOLD as an abstention, and produces relatively wide confidence intervals. It therefore cannot establish that heterogeneity improves or fails to improve predictive performance. The reported hit rates should not be interpreted as evidence of investment alpha, trading profitability, or economically significant forecasting ability.

VIII. CONCLUSION

This study presents a pre-registered empirical evaluation of how agreement and inference cost differ across LLM ensemble configurations of increasing cross-provider heterogeneity. Across a 54,000-call primary experiment, within-ensemble agreement decreases monotonically as provider diversity increases, with a primary HOM-to-HET contrast of $\Delta\kappa = 0.336$ (95% CI: $[0.304, 0.369]$). This reduction is accompanied by an approximately $4.4\times$ increase in inference cost, producing an explicit agreement–cost frontier for the evaluated configurations. A capability-matched control preserves approximately 94% of the primary contrast, which therefore remains substantial under approximate capability matching; provider nevertheless remains bundled with other model-specific characteristics and is not independently identified. We also introduce an independent-draw measurement protocol that controls dependence caused by shared sampling seeds. In a controlled pilot, shared per-run seeding inflated the estimated agreement contrast by approximately fourfold, demonstrating the importance of checking for residual within-run dependence and avoiding shared-seed artifacts in ensemble-agreement studies.

Cross-provider heterogeneity does not produce a statistically reliable improvement in average directional accuracy at the scale of this experiment. However, exploratory results indicate that homogeneous ensembles generate unanimous

incorrect directional decisions at approximately $1.84\times$ the rate of fully heterogeneous ensembles and concentrate more agents around incorrect majority decisions. These findings suggest that heterogeneity may provide value by reducing concentrated collective errors and preserving disagreement signals, although this interpretation requires confirmatory evaluation. Overall, the results show that model diversity can reduce output agreement and error correlation, but the practical value of that reduced agreement must be established against a downstream objective such as calibration, robustness, abstention quality, or collective-error reduction. Future work should evaluate the observed agreement ordering across additional tasks, ensemble compositions, and fully tier-matched provider configurations.

DATA AND CODE AVAILABILITY

To support reproduction, we release: (i) the immutable, read-only, SHA-256-stamped raw dataset (`runs.csv`), containing all 94,566 logged attempts—the 54,000 valid responses together with every unsuccessful request and its error—each with the per-call model, seed, prompt hash, timestamp, token usage, cost, and raw response; (ii) the pre-registration materials and the cryptographic freeze receipts for both the confirmatory study and the cross-domain replication; (iii) the deterministic, seeded analysis pipeline that regenerates every table and figure in this paper, together with a replication check; and (iv) the orchestration harness, including the HetSameTier control configuration. Code and data are available at https://github.com/samirrc2/price-of-decorrelation. A Code Ocean compute capsule enabling keys-free reproduction is available at https://doi.org/10.24433/CO.9524962.v2 (DOI: `10.24433/CO.9524962.v2`). The dataset SHA-256 hash and analysis commit hash are recorded in the supplementary manifest.

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USE OF GENERATIVE AI

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